

一、实验目的

1. 熟悉 Wireshark 的基本操作方法。
 2. 掌握抓取和分析网络数据包的基本技能。
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二、实验原理

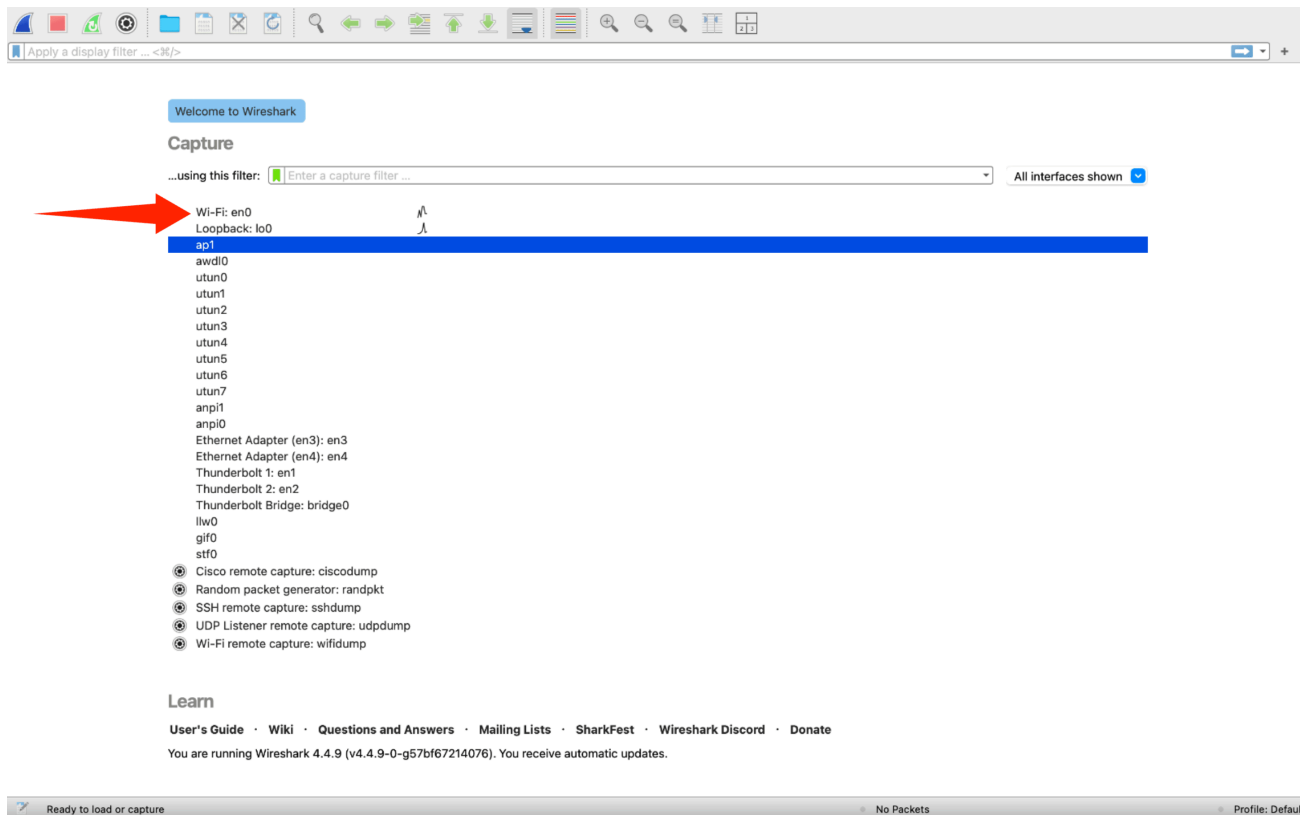
1. **Wireshark**：一种常用的网络协议分析工具，主要由两部分组成
 - **数据包捕获库**：接收计算机发送 / 接收的所有链路层帧
 - **数据包分析器**：解析并显示协议消息中的所有字段
 2. **协议封装**：HTTP 报文封装在 TCP 段内，TCP 段封装在 IP 数据报内，IP 数据报封装在以太网帧内。
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三、实验环境

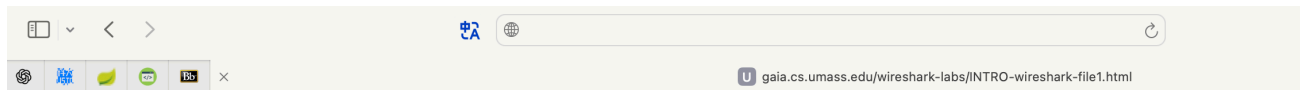
- 操作系统：macOS
 - 软件工具：Wireshark
 - 网络环境：无线网络
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四、实验步骤

1. 启动 Wireshark，选择合适的网络接口，即Wi-Fi en0，并开始抓包。



2. 打开浏览器，并访问:<http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html>



3. 停止抓包

No.	Time	Source	Destination	Protocol	Length	Info
5371	16:20:37.083229	202.38.64.56	100.64.173.157	DNS	130	Standard query response 0xb65b HTTPS gaia.cs.umass.edu SOA unix1.cs.umass.edu
5372	16:20:37.083231	202.38.64.56	100.64.173.157	DNS	93	Standard query response 0x0cb2 A gaia.cs.umass.edu A 128.119.245.12
5373	16:20:37.083232	202.38.64.56	100.64.173.157	DNS	130	Standard query response 0x9439 AAAA gaia.cs.umass.edu SOA unix1.cs.umass.edu
5374	16:20:37.085033	100.64.173.157	128.119.245.12	TCP	78	62952 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=1975005095 TSecr=0 SACK_PERM
5375	16:20:37.422056	128.119.245.12	100.64.173.157	TCP	74	80 → 62952 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1388 SACK_PERM TSval=2086128244 TSecr=
5376	16:20:37.422614	100.64.173.157	128.119.245.12	TCP	66	62952 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0 TSval=1975005433 TSecr=2086128244
5377	16:20:37.423672	100.64.173.157	128.119.245.12	TCP	78	62953 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=674479677 TSecr=0 SACK_PERM
5378	16:20:37.730078	128.119.245.12	100.64.173.157	TCP	74	80 → 62953 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1388 SACK_PERM TSval=2086128583 TSecr=
5379	16:20:37.731762	100.64.173.157	128.119.245.12	TCP	66	62953 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0 TSval=674479984 TSecr=2086128583
5380	16:20:37.731768	100.64.173.157	128.119.245.12	HTTP	498	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
5381	16:20:38.002236	2001:da8:d800:b749...	2a02:26f7:149:0:ac...	QUIC	92	Protected Payload (KP0)
5382	16:20:38.018219	128.119.245.12	100.64.173.157	TCP	66	80 → 62953 [ACK] Seq=1 Ack=433 Win=30080 Len=0 TSval=2086128891 TSecr=674479985
5383	16:20:38.018223	128.119.245.12	100.64.173.157	HTTP	504	HTTP/1.1 200 OK (text/html)
5384	16:20:38.019035	100.64.173.157	128.119.245.12	TCP	66	62953 → 80 [ACK] Seq=433 Ack=439 Win=130880 Len=0 TSval=674480272 TSecr=2086128891
5385	16:20:38.101455	2a02:26f7:149:0:ac...	2001:da8:d800:b749...	QUIC	94	Protected Payload (KP0)
5386	16:20:40.391024	121.204.244.17	100.64.173.157	TCP	124	13920 → 62116 [PSH, ACK] Seq=117 Ack=203 Win=10 Len=58 TSval=632675102 TSecr=3258303720
5387	16:20:40.391153	100.64.173.157	121.204.244.17	TCP	66	62116 → 13920 [ACK] Seq=203 Ack=175 Win=2048 Len=0 TSval=3258323688 TSecr=632675102
5388	16:20:40.391474	100.64.173.157	121.204.244.17	TCP	150	62116 → 13920 [PSH, ACK] Seq=203 Ack=175 Win=2048 Len=84 TSval=3258323688 TSecr=632675102
5389	16:20:40.437255	121.204.244.17	100.64.173.157	TCP	66	13920 → 62116 [ACK] Seq=175 Ack=287 Win=10 Len=0 TSval=632675192 TSecr=3258323688

> Frame 5380: 498 bytes on wire (3984 bits), 498 bytes captured (3984 bits) on interface e

> Ethernet II, Src: 12:1b:8c:50:f4:26 (12:1b:8c:50:f4:26), Dst: HuaweiTechno_8a:5d:45 (c8:

> Internet Protocol Version 4, Src: 100.64.173.157, Dst: 128.119.245.12

> Transmission Control Protocol, Src Port: 62953, Dst Port: 80, Seq: 1, Ack: 1, Len: 432

> Hypertext Transfer Protocol

0000 c8 33 e5 8a 5d 45 12 1b 8c 50 f4 26 08 00 45 00 ...3...E...P&...E...

0010 01 e4 00 00 40 00 40 06 b1 b2 64 40 ad 9d 80 77 ...@...@...d...w...

0020 f5 0c f5 e9 00 50 77 bb 5d 1a 44 fc c5 0b 80 18 ...Pw...D...

0030 08 04 d9 17 00 00 01 01 08 0a 28 33 bf 71 7c 57(3q)W...

0040 cb c7 47 45 54 20 2f 77 69 72 65 73 68 61 72 6b ...GET /w ireshark...

0050 2d 6c 61 62 73 2f 49 4e 54 52 4f 2d 77 69 72 65 ...-labs/IN TR0-wire...

0060 73 68 61 72 6b 2d 66 69 6c 65 31 2e 68 74 6d 6c ...shark-fi le1.html...

0070 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a ...HTTP/1.1: Host:

0080 20 67 61 69 61 2e 63 73 2e 75 6d 61 73 73 2e 65 ...gaia.cs .umass.e...

0090 64 75 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 ...du-User -Agent:

00a0 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 4d 61 63 ...Mozilla/ 5.0 (Mac...

00b0 69 6e 74 6f 73 68 3b 20 49 6e 74 65 6c 20 4d 61 ...intosh; Intel Ma...

00c0 63 20 4f 53 20 58 20 31 30 5f 31 35 5f 37 29 20 ...c OS X 1 0.15.7)

00d0 41 70 70 6c 65 57 65 62 4b 69 74 2f 36 30 35 2e ...AppleWeb Kit/605...

00e0 31 2e 31 35 20 28 4b 48 54 4d 4c 2c 20 6c 69 6b ...1.15 (KH TML, lik...

00f0 65 20 47 65 63 6b 6f 29 20 56 65 72 73 69 6f 6e ...e Gecko) Version...

0100 2f 31 38 2e 31 2e 31 20 53 61 66 61 72 69 2f 36 .../18.1.1 Safari/6...

0110 30 35 2e 31 2e 31 35 0d 0a 55 70 67 72 61 64 65 ...05.1.15- Upgrade...

0120 2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73 ...-Insecur e-Reques...

0130 74 73 3a 20 31 0d 0a 41 63 63 65 70 74 3a 20 74 ...ts: 1 A ccept: t...

0140 65 78 74 2f 68 74 6d 6c 2c 61 70 70 6c 69 63 61 ...ext/html ,applica...

0150 74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 ...tion/xht ml+xml;a...

0160 70 70 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 3b 71 ...pplicati on/xml;q...

0170 3d 30 2e 39 2c 2a 2f 2a 3b 71 3d 30 2e 38 0d 0a ...=0.9,*/; ;q=0.8...

0180 41 63 63 65 70 74 2d 4c 61 6e 67 75 61 67 65 3a ...Accept-L anguage:

0190 20 7a 68 2d 43 4e 2c 7a 68 2d 48 61 6e 73 3b 71 ...zh-CN,z h-Hans;q...

01a0 3d 30 2e 39 0d 0a 50 72 69 6f 72 69 74 79 3a 20 ...=0.9-Pr iority;

01b0 75 3d 30 2c 20 69 0d 0a 41 63 63 65 70 74 2d 45 ...u=0, i... Accept-E...

01c0 6e 63 6f 64 69 6e 67 3a 20 67 7a 69 70 2c 20 64 ...ncoding: gzip, d...

wireshark_Wi-Fi312RC3.pcapng

Packets: 13482

Profile: Default

4. 筛选出其中有关于http的记录

No.	Time	Source	Destination	Protocol	Length	Info
5380	16:20:37.731768	100.64.173.157	128.119.245.12	HTTP	498	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
5383	16:20:38.018223	128.119.245.12	100.64.173.157	HTTP	504	HTTP/1.1 200 OK (text/html)
8231	16:20:56.752615	100.64.173.157	23.66.33.52	HTTP	424	GET /ME8wTBLMEkwzrAHBgUrDgMCgGQU0dKLCf4dGbZfs%2FE0jy08BFlcQ5UEF41VCAYIebjbuYP%2Bvq5Eu0Gf48:
8244	16:20:56.815340	23.66.33.52	100.64.173.157	OCSP	792	Response

> Frame 5380: 498 bytes on wire (3984 bits), 498 bytes captured (3984 bits) on interface e

> Ethernet II, Src: 12:1b:8c:50:f4:26 (12:1b:8c:50:f4:26), Dst: HuaweiTechno_8a:5d:45 (c8:

> Internet Protocol Version 4, Src: 100.64.173.157, Dst: 128.119.245.12

> Transmission Control Protocol, Src Port: 62953, Dst Port: 80, Seq: 1, Ack: 1, Len: 432

> Hypertext Transfer Protocol

0000 c8 33 e5 8a 5d 45 12 1b 8c 50 f4 26 08 00 45 00 ...3...E...P&...E...

0010 01 e4 00 00 40 00 40 06 b1 b2 64 40 ad 9d 80 77 ...@...@...d...w...

0020 f5 0c f5 e9 00 50 77 bb 5d 1a 44 fc c5 0b 80 18 ...Pw...D...

0030 08 04 d9 17 00 00 01 01 08 0a 28 33 bf 71 7c 57(3q)W...

0040 cb c7 47 45 54 20 2f 77 69 72 65 73 68 61 72 6b ...GET /w ireshark...

0050 2d 6c 61 62 73 2f 49 4e 54 52 4f 2d 77 69 72 65 ...-labs/IN TR0-wire...

0060 73 68 61 72 6b 2d 66 69 6c 65 31 2e 68 74 6d 6c ...shark-fi le1.html...

0070 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a ...HTTP/1.1: Host:

0080 20 67 61 69 61 2e 63 73 2e 75 6d 61 73 73 2e 65 ...gaia.cs .umass.e...

0090 64 75 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 ...du-User -Agent:

00a0 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 4d 61 63 ...Mozilla/ 5.0 (Mac...

00b0 69 6e 74 6f 73 68 3b 20 49 6e 74 65 6c 20 4d 61 ...intosh; Intel Ma...

00c0 63 20 4f 53 20 58 20 31 30 5f 31 35 5f 37 29 20 ...c OS X 1 0.15.7)

00d0 41 70 70 6c 65 57 65 62 4b 69 74 2f 36 30 35 2e ...AppleWeb Kit/605...

00e0 31 2e 31 35 20 28 4b 48 54 4d 4c 2c 20 6c 69 6b ...1.15 (KH TML, lik...

00f0 65 20 47 65 63 6b 6f 29 20 56 65 72 73 69 6f 6e ...e Gecko) Version...

0100 2f 31 38 2e 31 2e 31 20 53 61 66 61 72 69 2f 36 .../18.1.1 Safari/6...

0110 30 35 2e 31 2e 31 35 0d 0a 55 70 67 72 61 64 65 ...05.1.15- Upgrade...

0120 2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73 ...-Insecur e-Reques...

0130 74 73 3a 20 31 0d 0a 41 63 63 65 70 74 3a 20 74 ...ts: 1 A ccept: t...

0140 65 78 74 2f 68 74 6d 6c 2c 61 70 70 6c 69 63 61 ...ext/html ,applica...

0150 74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 ...tion/xht ml+xml;a...

0160 70 70 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 3b 71 ...pplicati on/xml;q...

0170 3d 30 2e 39 2c 2a 2f 2a 3b 71 3d 30 2e 38 0d 0a ...=0.9,*/; ;q=0.8...

0180 41 63 63 65 70 74 2d 4c 61 6e 67 75 61 67 65 3a ...Accept-L anguage:

0190 20 7a 68 2d 43 4e 2c 7a 68 2d 48 61 6e 73 3b 71 ...zh-CN,z h-Hans;q...

01a0 3d 30 2e 39 0d 0a 50 72 69 6f 72 69 74 79 3a 20 ...=0.9-Pr iority;

01b0 75 3d 30 2c 20 69 0d 0a 41 63 63 65 70 74 2d 45 ...u=0, i... Accept-E...

01c0 6e 63 6f 64 69 6e 67 3a 20 67 7a 69 70 2c 20 64 ...ncoding: gzip, d...

Hypertext Transfer Protocol: Protocol

Packets: 15214 · Displayed: 4 (0.0%) · Dropped: 0 (0.0%)

Profile: Default

五、实验问题

Q1

通过观察抓包得到的数据中的protocol一栏，观察到了包括但不限于如下三种协议：

- UDP
- QUIC
- DNS

Q2

No.	Time	Source	Destination	Protocol	Length	Info
32	19:40:32.286363	100.64.166.18	128.119.245.12	HTTP	583	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
35	19:40:32.571974	128.119.245.12	100.64.166.18	HTTP	305	HTTP/1.1 304 Not Modified

通过Time栏可看出，由发送请求到服务器返回内容，共花费了 0.285611s

Q3

由两条记录的Source以及Destination可以看出：

- gaia.cs.umass.edu地址：128.119.245.12
- 我的地址：100.64.166.18

Q4

两条关于http的记录的信息如下（pdf一并在上传的压缩包中给出）

/var/folders/_4/h7b2_46x4zx4v2c1w0zwt4b00000gn/T/wireshark_Wi-FiSBHYC3.pcapng 80 total packets, 1 shown

```
No.      Time                Source                Destination            Protocol Length Info
  32  19:40:32.286363    100.64.166.18         128.119.245.12        HTTP      583    GET /
wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
Frame 32: 583 bytes on wire (4664 bits), 583 bytes captured (4664 bits) on interface en0, id 0
Ethernet II, Src: 12:1b:8c:50:f4:26 (12:1b:8c:50:f4:26), Dst: HuaweiTechno_8a:5d:45
(c8:33:e5:8a:5d:45)
  Destination: HuaweiTechno_8a:5d:45 (c8:33:e5:8a:5d:45)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0. .... = IG bit: Individual address (unicast)
  Source: 12:1b:8c:50:f4:26 (12:1b:8c:50:f4:26)
    .... 1. .... = LG bit: Locally administered address (this is NOT the
factory default)
    .... 0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]
Internet Protocol Version 4, Src: 100.64.166.18, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 569
  Identification: 0x0000 (0)
  010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: TCP (6)
  Header Checksum: 0xb8e8 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 100.64.166.18
  Destination Address: 128.119.245.12
  [Stream index: 3]
Transmission Control Protocol, Src Port: 64666, Dst Port: 80, Seq: 1, Ack: 1, Len: 517
Hypertext Transfer Protocol
```

/var/folders/_4/h7b2_46x4zx4v2clw0zwt4b00000gn/T/wireshark_Wi-FiSBHYC3.pcapng 80 total packets, 1 shown

```
No.      Time                Source                Destination           Protocol Length Info
  35  19:40:32.571974      128.119.245.12        100.64.166.18         HTTP      305      HTTP/1.1
304 Not Modified
Frame 35: 305 bytes on wire (2440 bits), 305 bytes captured (2440 bits) on interface en0, id 0
Ethernet II, Src: HuaweiTechno_8a:5d:45 (c8:33:e5:8a:5d:45), Dst: 12:1b:8c:50:f4:26 (12:1b:8c:
50:f4:26)
    Destination: 12:1b:8c:50:f4:26 (12:1b:8c:50:f4:26)
        .... 1. .... = LG bit: Locally administered address (this is NOT the
factory default)
        .... 0 .... = IG bit: Individual address (unicast)
    Source: HuaweiTechno_8a:5d:45 (c8:33:e5:8a:5d:45)
        .... 0. .... = LG bit: Globally unique address (factory default)
        .... 0 .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 0]
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 100.64.166.18
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 291
    Identification: 0x55af (21935)
    010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 27
    Protocol: TCP (6)
    Header Checksum: 0x894f [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 128.119.245.12
    Destination Address: 100.64.166.18
    [Stream index: 3]
Transmission Control Protocol, Src Port: 80, Dst Port: 64666, Seq: 1, Ack: 518, Len: 239
Hypertext Transfer Protocol
```

- 从该打印内容中可获取如下有用信息：

- **HTTP GET 报文 (Frame 32)**

- 时间戳：19:40:32.286363
- 源 IP 地址：100.64.166.18 (本机)
- 目的 IP 地址：128.119.245.12 (服务器 gaia.cs.umass.edu)
- 源端口：64666
- 目的端口：80 (HTTP 默认端口)
- 传输层协议：TCP
- 应用层协议：HTTP
- 请求内容：GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
- 报文长度：583 字节

说明本地主机向目标服务器发送了一个网页请求，访问指定的实验文件。

- **HTTP 响应报文 (Frame 35)**

- 时间戳：19:40:32.571974
- 源 IP 地址：128.119.245.12 (服务器)
- 目的 IP 地址：100.64.166.18 (本机)
- 源端口：80
- 目的端口：64666
- 传输层协议：TCP

- 应用层协议: HTTP
- 响应内容: HTTP/1.1 304 Not Modified
- 报文长度: 305 字节

服务器返回了 **304 Not Modified** 状态码, 表示客户端缓存的内容仍然有效, 因此无需再次传输文件内容。

六、实验结论

1. 在本次试验中, 我掌握了 Wireshark 的基本使用方法, 例如选择接口、抓包、过滤和查看协议内容。
2. 实验结果表明, 网络传输过程包含多种协议协同工作, 并不是仅有http或者https, 并且用户简单的网页访问背后可能涉及复杂的协议交互。